Assessment Coversheet

Qualification details			
Training Package Code and Title:	**ICT - Information and Communications Technology (Release 8.1)**		
Qualification National Code and Title:	**ICT40120 - Certificate IV in Information Technology (Release 4)**	**State code:**	**BFF9**

Assessment Title	**AT2.3 Project (Build & Test a GUI)**		
Unit National Code & Title	ICTPRG440 - Apply introductory programming skills in different languages		
	ICTPRG437 - Build a user interface		
	ICTICT449 - Use version control systems in development environments		
Date Due	**Session 14**	**Date Received**	

Student Name	John Richardson	**Student ID**	M196909
Student Declaration	I declare that the evidence submitted is my own work: John Richardson		

Assessor Name	Tirtha Ranjeet			
Assessment Decision	☐ Satisfactory		☐ Not Yet Satisfactory	
Assessor Signature			**Date**	
Is student eligible for reassessment (Re-sit)?	☐ No	☐ Yes	**Reassessment Date:**	

Feedback to student
Via Blackboard (LMS) – Please check [Grade] section.
Feedback from student
Via Blackboard (LMS) – Please use [Comment] section during submission.

Student signature		**Date**	

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Assessment Instructions

TO THE ASSESSOR

Type of Assessment	Project
Duration of Assessment	5 Class Sessions (10 - 14)
Location of Assessment	Classroom
Conditions	Assessor to ensure that the noise levels, natural interactions and time variances are maintained as it would in the be in the Cyber Security industry. Learners are required to complete the required tasks in class and submit the required documentation electronically via Blackboard
Elements and Criteria	As detailed in the assessment plan You are required to make sure that all students meet the elements, performance criteria and oral communication items as outlined in the provided checklist.

TO THE STUDENT

Purpose of Assessment	You are required to show you can: ICTPRG440 - Apply introductory programming skills in different languages ICTPRG437 - Build a user interface ICTICT449 - Use version control systems in development environments You are required to meet the elements, performance criteria and oral communication items as outlined in the provided checklist.
Allowable Materials	Blackboard (Topic by topic) will include the following: Weekly Readings, Class notes, and Weekly Activities.
Required Resources	Portable hard-drive (supplied by you) Computer with: • Internet Access • Microsoft Office (or Office 365) • Access to Learning Management System (Blackboard) • Web links and example code can be downloaded from Blackboard Learning Management System • Desktop Computer with Windows Operating System • Appropriate IDE for the programming language • Appropriate Framework (e.g. .NET 9.0 for the C# language) • Programming standards and guidelines found in the assessment tab of Blackboard.

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Reasonable Adjustment	In some circumstances, adjustments to assessments may be made for you. If you require support for literacy and numeracy issues; support for hearing, sight or mobility issues; change to assessment times/venues; use of special or adaptive technology; considerations relating to age, gender and cultural beliefs; format of assessment materials; or presence of a scribe you need to inform your lecturer.
Assessment Submission	All questions and activities must be attempted. Use of research tools and peers in formulating answers are acceptable – but work submitted must be your own work. Final project documentation is to be uploaded to the appropriate area in the Blackboard course created for this unit. If you are marked as NYS (Not Yet Satisfactory) on your first attempt, you will be provided with another opportunity to re-attempt the assessment.
Project contents	This project consists of the following tasks: • Create a graphical user interface (GUI) application • Use a VCS • Debug the application • Test the application • Demonstrate the application • Obtain sign-off

Task 3: Build and Test GUI, use VCS

Task 2 has the following accompanying documents and must be read in conjunction with them, they can be found in the Assessments tab on Black board. They are:

- C#One_AT2_ProjectBrief.pdf
- C#One_AT2.3_RoW.docx
- CITE-MS Programming Standards and Guidelines

There are two parts to this task. You will turn your prototype into a working GUI with full functionality, record tests and use your VCS.

Part 1

Activity A: Create GUI Application and use VCS

Your GUI project needs to include the following:

- A new branch in your chosen VCS for the GUI project
- Appropriate use of variables
- Comments as per the supplied organisational standards as well as XML commenting
- Sequence, selection and iteration constructs
- External library functions
- At least two data structures (e.g array, list, dictionary)
- Use standard algorithms to perform the following on the data structures:
    - Sequential search
    - Binary search
    - Insertion
    - Deletion
- Use the staging area of your chosen VCS appropriately (add and commit files)
- Commit files to local repository

Evidence will be provided with the complete application project as well as the following screenshots pasted into the appropriate section of your record of work document (**Activity A – VCS Evidence**) - see next page

Activity A: VCS Evidence

i. **Add and commit files locally**
Make a meaningful change to your project (e.g. update the README file, change a source file), add changed files to the changing area according to the serviced provider's normal workflow. Create one or more commits that include these staged files using clear and descriptive messages that follow the organisational guidelines

ii. Check version history and confirm status
Open the commit history for your repository and identify the initial commit and the most recent commit. For each of these two commits record the following:

- The commit ID or reference
- The date and time
- A short description of what changed

iii. Respond to a pull request on the remote repository
Navigate to your repository on the web interface of your chosen platform. Select the base branch then select the compare branch. Add a clear title and description explaining the purpose of your changes then submit the pull request. Take appropriate action to respond to the request locally.

iv. Retrieve and merge remote commits locally

- Create a new commit directly on the remote repository then, on your local machine, modify a file in the project
- Stage and commit this change locally with a clear commit message (don't pull from the remote yet.
- Retrieve and merge the remote commit
- Confirm the final local state
    - Show that the local branch shows both your local and remote changes

Part 2: Debug and Test

Activity A: Debug

Using both a stand-alone debugger and the Visual Studio built in debugger provide evidence of debugging an issue in your application.

Provide evidence in the for of a screen recordings or a separate document of detailed screenshots .

Name the file:

- **StudentID_AT2.3_Debug.mp4 or StudentID_AT2.3_Debug.pdf**

Activity B: Test

Using the supplied test plan template in your record of work document thoroughly test your application.

Activity D: Demonstration and Sign-Off

Demonstrate your application to your supervisor (Lecturer). Obtain feedback and make any changes if required.
Have your supervisor complete the sign-off form in your Record of Work document once complete.

AT2.3 Submission

StudentID_AT2.1_RoW.docx

StudentID_AT2.3_Debug.mp4 or StudentID_At2.3_Debug.pdf

StudentID_AT2.3_GUI.zip (complete solution folder)